Faculty of Engineering and Technology

Electrical and Computer Engineering Department

Machine Learning and Data Science - ENCS5341

Assignment #1

Data Preprocessing & Exploratory Data Analysis (EDA)

Prepared by

Name :Ahmad karmi **ID:1220003**

Name: Nabeel Sawafta **ID:1220869**

Instructor: Yazan Abu Farha

Section:2

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preprocessing steps

First, we loaded the dataset into a pandas DataFrame and examined its structure and content using commands like df.head(), df.info(), and df.describe(). To identify any missing data, we used df.isnull().sum() to count the missing values in each column.

For handling missing values, we applied different strategies depending on the data type: for continuous numerical attributes such as Age, Income, and Tenure, missing values were replaced with the median of the respective column, while for discrete numerical attributes like SupportCalls, missing values were filled with the mode, ensuring that the dataset was complete and ready for analysis.

Next, we investigated the presence of outliers using boxplots, which visually highlight extreme values in each numeric attribute. For columns with outliers, we calculated the first quartile (Q1), third quartile (Q3), interquartile range (IQR), and the corresponding upper and lower limits based on the standard IQR method (Q1 - 1.5*IQR, Q3 + 1.5*IQR).

- For the Age attribute, we chose to remove the outliers entirely, which in our dataset corresponded to two individuals under the age of 20.
- For Income, outliers were replaced with the mean, which preserves the dataset size while reducing the effect of extreme values.
- For SupportCalls, outliers were replaced with the median, as this attribute is discrete and the median provides a more robust central tendency measure for count data.

Once the data was cleaned, we performed normalization using Min-Max scaling from sklearn.preprocessing to rescale all numeric features to the range [0,1], ensuring that variables with different units or scales were comparable and suitable for further analysis.

For exploratory data analysis (EDA), we employed Matplotlib and Seaborn libraries to visualize the data distributions and relationships. Histograms were used to understand the spread and distribution of numeric attributes, while bar plots allowed us to examine the distribution of categorical variables like Gender and ProductType. Boxplots were used again after outlier handling to verify the effectiveness of our cleaning procedures.

Additionally, we explored the relationship between numeric features and the target variable ChurnStatus using boxplots grouped by churn, and we investigated the interactions between categorical variables and churn using grouped bar plots. For a comprehensive view of correlations between numerical features, a heatmap was plotted.

Finally, we leveraged Seaborn's pairplot to visualize pairwise relationships between numeric features, providing insights into patterns, clusters, and potential correlations in the dataset.

Overall, the preprocessing and visualization steps ensured that the data was clean, normalized, and fully explored, providing a strong foundation for any subsequent analysis or modeling tasks.

justifications for the preprocessing decisions

1. Handling Missing Values

In the dataset, we observed missing values in both **continuous** and **discrete** columns. To handle them, we chose **imputation** rather than deletion, and our strategy differed based on the type of data:

- **Continuous numerical attributes** (Age, Income, Tenure):
  We replaced missing values with the **median** of each column.
  Justification:
  - The median is robust to outliers, unlike the mean.
  - It preserves the central tendency of the data without being skewed by extreme values.
  - Deleting rows with missing values could reduce dataset size unnecessarily, which is undesirable if missing values are few.
- **Discrete numerical attributes** (SupportCalls):
  We replaced missing values with the **mode** (most frequent value).
  Justification:
  - SupportCalls is a count variable; using the mode ensures that we fill missing entries with a realistic and frequent value.
  - This prevents introducing artificial fractional values that wouldn't make sense for counts.

Overall, imputation allows us to retain the full dataset, maintain statistical integrity, and avoid losing valuable information.

2. Handling Outliers

We identified outliers in numeric columns using **boxplots** and the **IQR method**. Our handling strategy differed for each column based on the data type and business context:

- **Age:**
  Outliers were **removed** from the dataset.
  Justification:
  - Extreme values (e.g., individuals under 20) are unrealistic in the business context.
  - Removing these outliers ensures that statistical measures like mean and standard deviation are not distorted.
- **Income:**
  Outliers were **replaced with the mean**.
  Justification:
  - Extremely high or low incomes could skew the data.
  - Replacing with the mean reduces the impact of these extremes while preserving the dataset size.
  - The mean is appropriate here because income is continuous and normally distributed in most cases.
- **SupportCalls:**
  Outliers were **replaced with the median**.
  Justification:
  - SupportCalls is a discrete count variable. Extreme values may be legitimate but can distort analysis.
  - The median provides a robust central value that is less sensitive to extreme counts, preserving realistic behavior without removing data.

Overall, our approach ensures that the dataset remains **clean, realistic, and suitable for analysis**, while minimizing information loss and bias from extreme values.

Visualizations

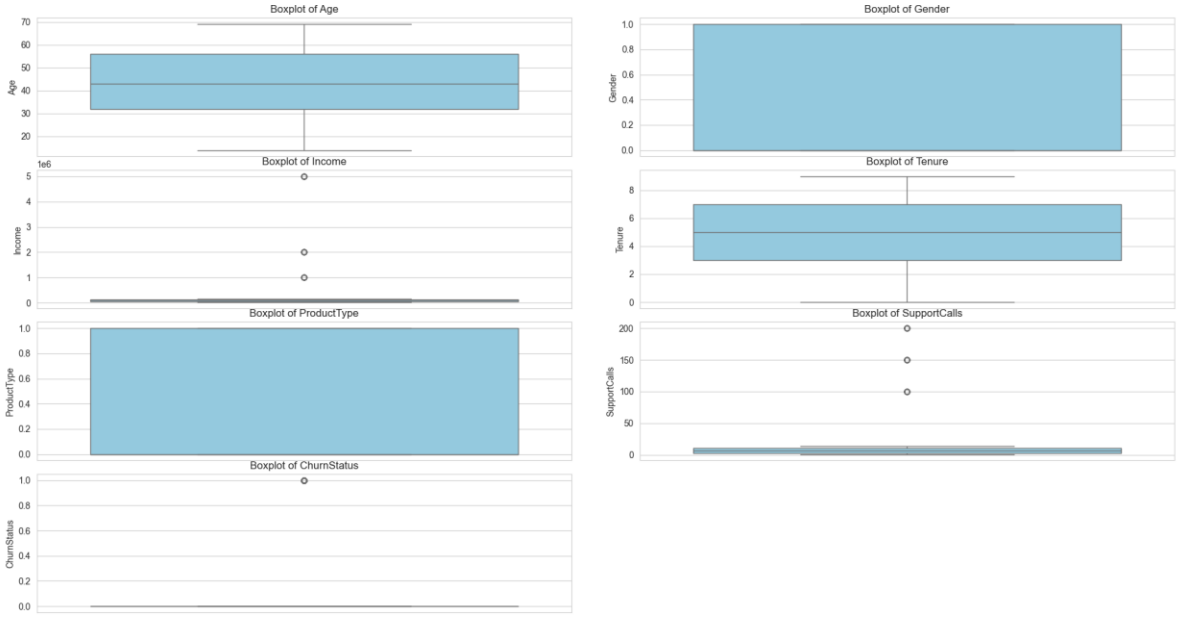

Figure 1:box plots before handling the outliers

We can observe the presence of an outlier above the upper limit in both the Product Type and Support Calls variables.

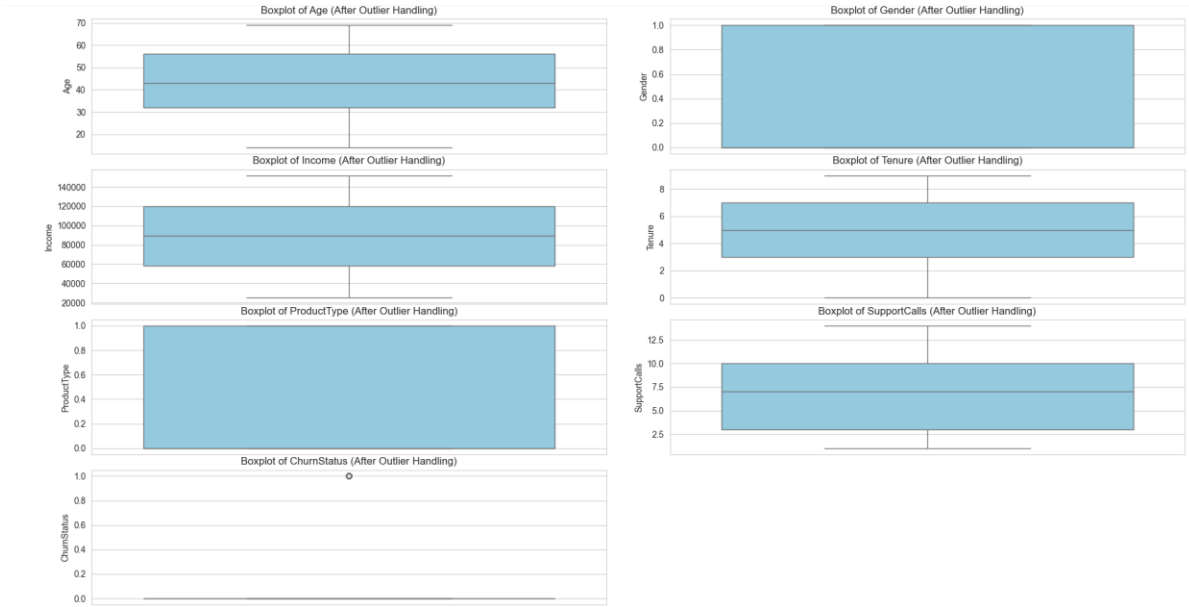

Figure 2:box plots after handling the outliers

We can observe that there are no remaining outliers in these variables after the data was processed.

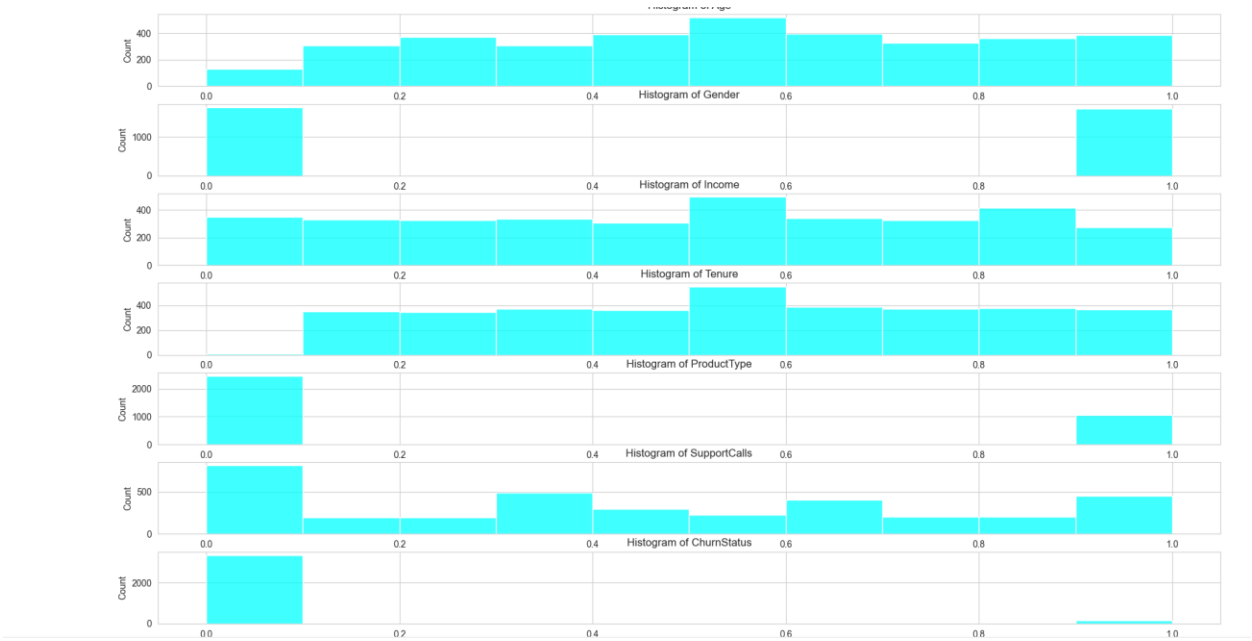

Figure 3:histograms plot

Age:

The distribution is relatively uniform across the entire range, indicating that customers span a broad spectrum of age groups without a strong concentration in any specific segment.

Key Insight: The customer base includes a well-balanced mix of both younger and older individuals.

Gender:

The histogram shows two distinct peaks at 0 and 1, likely corresponding to two gender categories (e.g.,0=Male,1=Female),with nearly equal frequencies.

Key Insight: Gender representation among customers is balanced.

Income:

The values are more concentrated at both the lower and upper ends of the range, likely due to the effects of normalization.

Key Insight: The customer base includes both lower- and higher-income individuals, supporting opportunities for income-based segmentation and targeted marketing.

Tenure:

Data points are distributed across the full range, indicating a mix of both newer and long-standing customers.

Key Insight: The company serves a healthy blend of new and loyal customers, providing valuable opportunities for churn prediction and retention strategies.

ProductType:

The histogram features spikes at specific intervals, suggesting a categorical variable with certain product types being more prevalent than others.

Key Insight: Some products are more popular than others, which can inform product development and promotional strategies.

SupportCalls:

Most customers are concentrated at the lower end of the scale, indicating that few support calls are typical. A smaller number of customers make frequent support calls.

Key Insight: While the majority of customers rarely contact support, a small group requires frequent assistance, potentially signaling dissatisfaction or product-related issues.

ChurnStatus:

Two clear peaks at 0 and 1 reflect the binary nature of the churn variable, with more customers remaining than leaving.

Key Insight: Although most customers stay with the company, there is a significant proportion who churn, emphasizing the importance of churn prevention efforts.

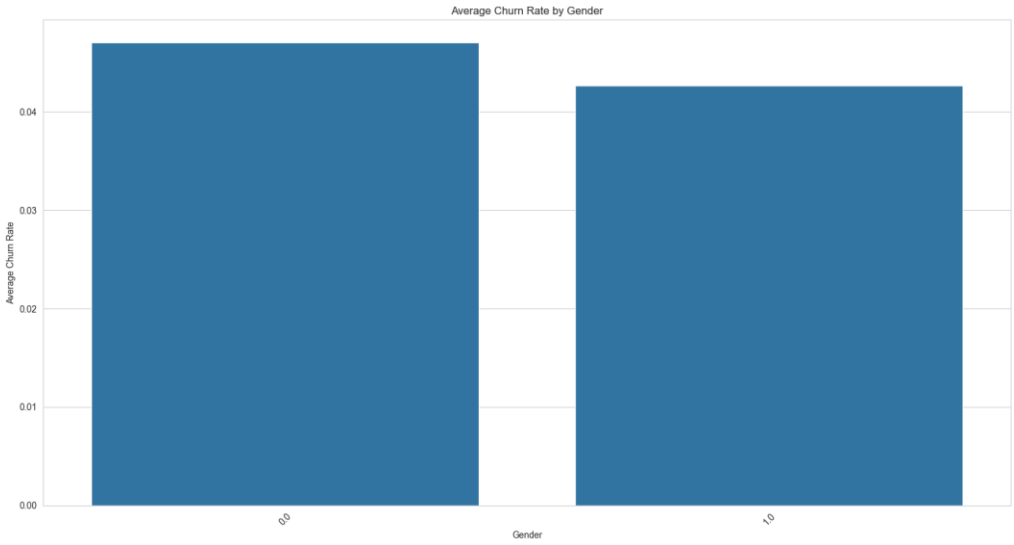

Figure 4: distribution of categorical variables using bar plots for gender

Explanation:
The bar chart illustrates the average churn rates for two gender categories (0:Male and 1:Female). Customers in Gender 0 exhibit a slightly higher average churn rate compared to those in Gender 1. This suggests that individuals in the first gender group are somewhat more likely to leave the service.

Key Insight:
While there is a minor difference in churn rates between the two gender groups—with Gender 0 showing a slightly higher propensity to churn—the variation is relatively small. This indicates that gender alone may not be a strong or decisive factor in predicting customer churn.

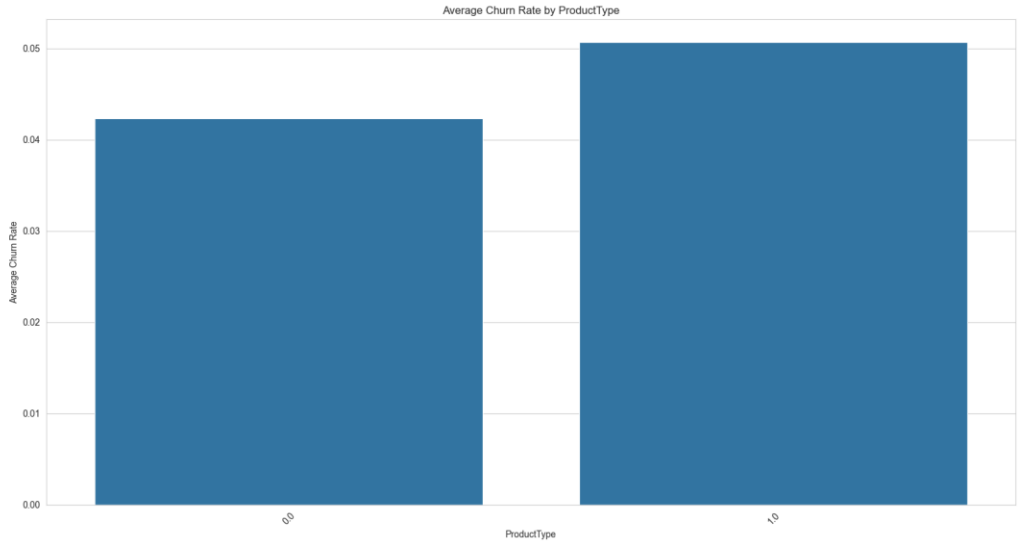

Figure 5:distribution of categorical variables using bar plots for product type

Explanation:

This bar chart compares the average churn rate between customers subscribed to different product types — 0 (Basic) and 1 (Premium). The churn rate is noticeably higher among customers with the Premium product compared to those with the Basic product.

Key Finding:

Customers using the Premium product are slightly more likely to churn than those with the Basic product. This could indicate potential dissatisfaction with the Premium service, pricing concerns, or unmet expectations, highlighting the need for deeper analysis into Premium customer experiences.

Figure 6: relationships between numerical features using box plots

1. Age vs ChurnStatus

- **Observation:** The distributions of age for churned and non-churned customers are quite similar and spread across the entire range.
- **Insight:** Age doesn't appear to have a strong influence on churn; customers of all ages churn at roughly similar rates.

2. Gender vs ChurnStatus

- **Observation:** Both genders (0 and 1) are almost equally represented in both churned and non-churned groups.
- **Insight:** Gender does not show a notable impact on churn. The distribution remains balanced across churn status.

3. Income vs ChurnStatus

- **Observation:** While most values are evenly distributed, there are a few outliers in the churned group.
- **Insight:** Income may have a slight relationship with churn, but overall distributions look quite similar. However, outliers in the churned group may warrant further investigation.

4. Tenure vs ChurnStatus

- **Observation:** Churned customers tend to have shorter tenures, while non-churned customers show a broader spread across the full tenure range.
- **Insight: Tenure is a strong indicator of churn.** Newer customers are more likely to churn, while longer-tenured customers are more likely to stay.

5. ProductType vs ChurnStatus

- **Observation:** The distributions are very similar across churn statuses.
- **Insight:** Product type does not appear to significantly influence churn in this dataset.

6. SupportCalls vs ChurnStatus

- **Observation:** Customers who churned tend to have slightly higher support call frequencies. The boxplot for churned users is shifted slightly upward, and the range is wider.
- **Insight: Higher support call frequency is moderately associated with churn**, possibly indicating dissatisfaction or unresolved issues.

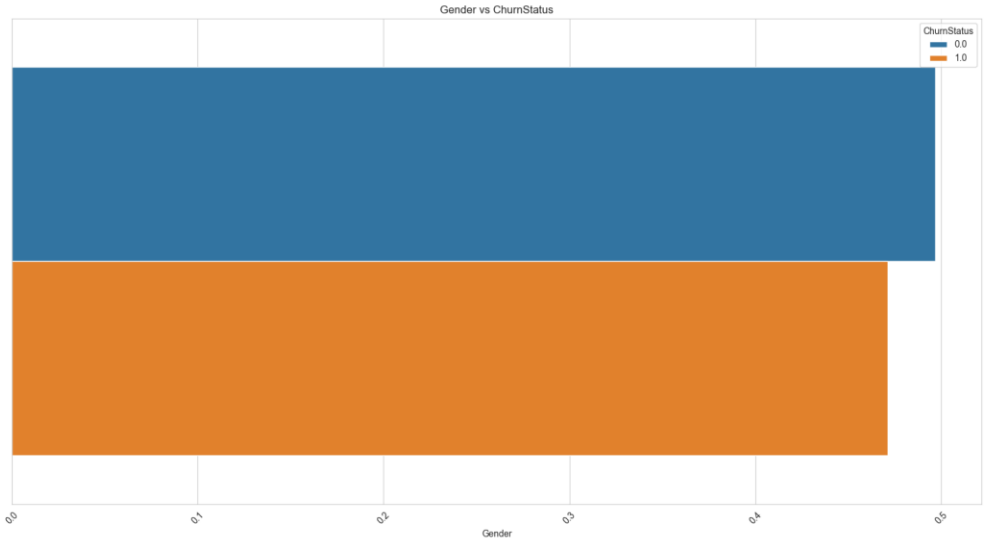

Figure 7:relationships between categorical variables using bar plots for gender

Here, we can observe that a higher proportion of males churned compared to females.

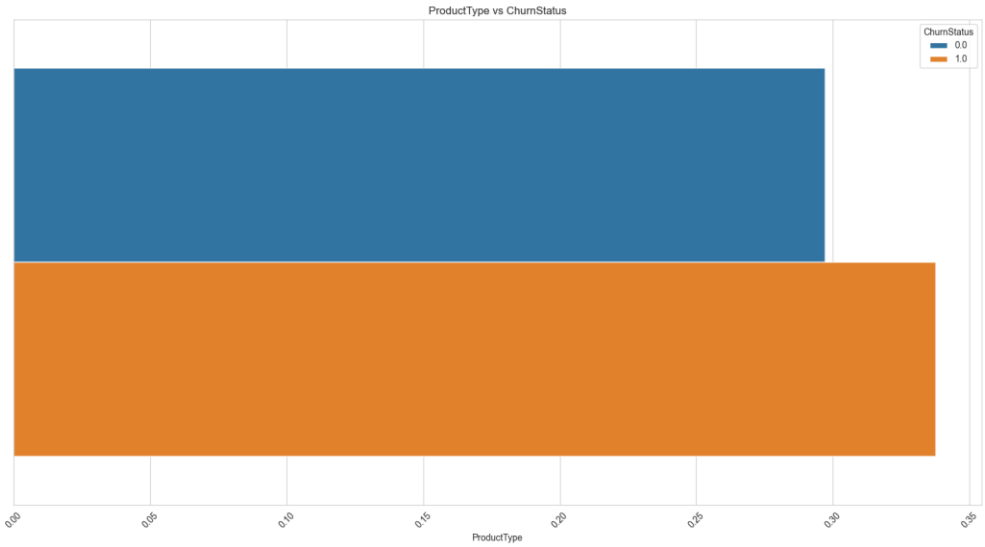

Figure 8:relationships between categorical variables using bar plots for product type

Customers subscribed to Prime churned more than those subscribed to Basic.

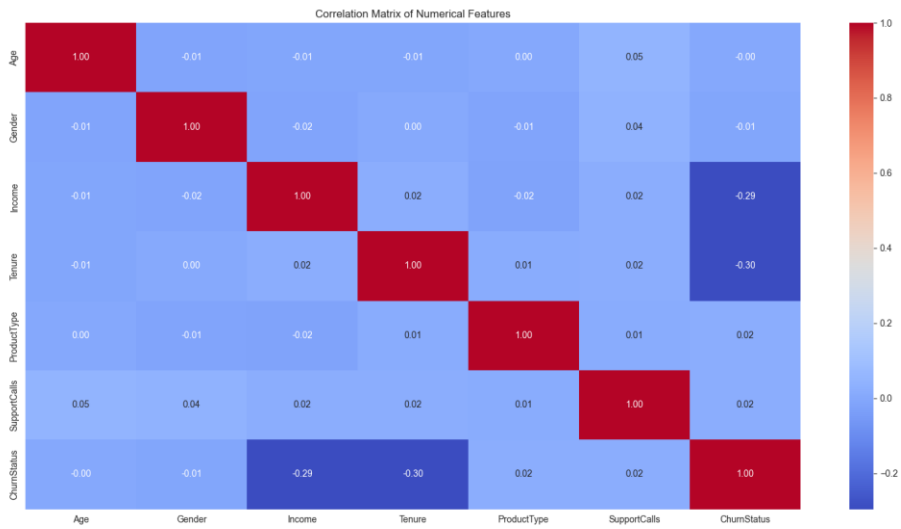

Figure 9:correlation matrix

This is a correlation matrix that shows the relationships between all attributes. Values close to 1 indicate a strong positive correlation (as one increases, the other also increases), values near 0 indicate little or no correlation, and values close to -1 indicate a strong negative correlation (as one increases, the other decreases).

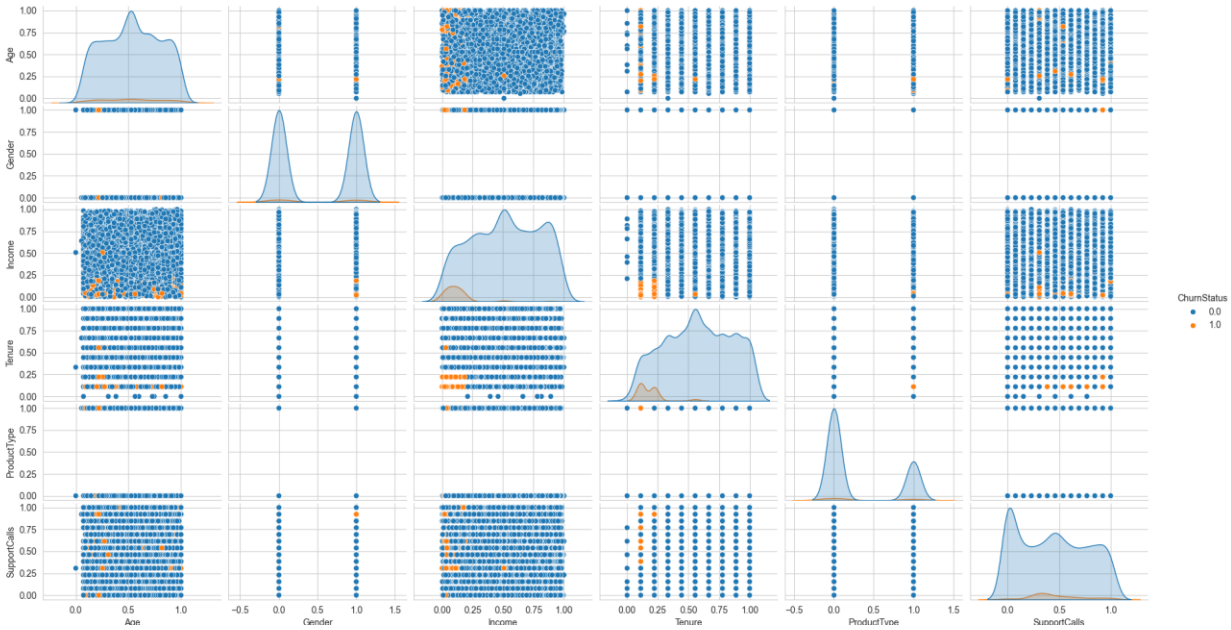

Figure 10:pair plots

The image is a pair plot illustrating the relationships between several normalized features—Age, Gender, Income, Tenure, ProductType, and SupportCalls—colored by ChurnStatus (0 for

retained customers, 1 for churned). The diagonal plots display the distribution of each variable, while the scatter plots highlight correlations between them. Orange points (churned customers) are mostly concentrated in the lower ranges of Income, Tenure, and SupportCalls, indicating that these features may be linked to higher churn. This plot helps uncover patterns and potential predictors of customer churn in the dataset.

Conclusion

The EDA indicates that customer churn is most strongly linked to tenure, support call frequency, and product type. Newer customers, those who contact support more frequently, and Premium subscribers exhibit higher churn rates, whereas age, gender, and income have minimal impact. While most customers remain with the company, a significant portion churns, emphasizing the importance of targeted retention strategies. Tenure emerges as a key predictor, suggesting that effective onboarding and early engagement programs could help retain new customers. Similarly, frequent support interactions may signal dissatisfaction or unresolved issues, so proactively addressing these cases can reduce churn. Premium subscribers also show a slightly higher propensity to leave, potentially due to unmet expectations or pricing concerns, highlighting the need to enhance the Premium experience. Focusing on these actionable factors—tenure, support interactions, and product experience—can help identify at-risk customers early and guide interventions that improve loyalty and retention.